

№ 1

 $f(x) = \frac{1}{\sqrt{e}}$ вычислить с точностью до $\delta = 0,001$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}; \quad x = -\frac{1}{3}$$

$$e^{-\frac{1}{3}} = 1 + \frac{-\frac{1}{3}}{1!} + \frac{(-\frac{1}{3})^2}{2!} + \frac{(-\frac{1}{3})^3}{3!} + \frac{(-\frac{1}{3})^4}{4!}$$

$$|-\frac{1}{3}| \approx 0,333...; \quad |\frac{1}{18}| \approx 0,0555...; \quad |-\frac{1}{162}| \approx 0,00617...; \quad |\frac{1}{1944}| \approx 0,0005144 < \delta$$

$$\frac{1}{\sqrt{e}} = 1 - \frac{1}{3} + \frac{1}{18} - \frac{1}{162} \approx 0,71605$$

№ 2

 $\sin 49^\circ$ вычислить с точностью до $\delta = 0,000001$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$x = 49^\circ = \frac{49\pi}{180}$$

$$\sin 49^\circ = \frac{49\pi}{180} - \frac{(\frac{49\pi}{180})^3}{3!} + \frac{(\frac{49\pi}{180})^5}{5!} - \frac{(\frac{49\pi}{180})^7}{7!} + \frac{(\frac{49\pi}{180})^9}{9!} + \dots$$

$$\frac{49\pi}{180} \approx 0,855...; \quad \left| -\frac{(\frac{49\pi}{180})^3}{3!} \right| \approx 0,104; \quad \left| \frac{(\frac{49\pi}{180})^5}{5!} \right| \approx 0,0038; \quad \left| -\frac{(\frac{49\pi}{180})^7}{7!} \right| \approx 0,0000664; \quad \left| \frac{(\frac{49\pi}{180})^9}{9!} \right| \approx 0,000000674$$

$$\frac{(\frac{49\pi}{180})^9}{9!} < \delta;$$

$$\sin 49^\circ = \frac{49\pi}{180} - \frac{(\frac{49\pi}{180})^3}{3!} + \frac{(\frac{49\pi}{180})^5}{5!} - \frac{(\frac{49\pi}{180})^7}{7!} \approx 0,7547089$$

№ 3: Вычислить предел.

$$\lim_{x \rightarrow 0} \frac{x - \arctg(x)}{x^3}$$

$$\arctg(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + \frac{x^{(2n-1)}}{(2n-1)}^{n-1}$$

$$x - \arctg(x) = x - \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right) = \frac{x^3}{3} - \frac{x^5}{5} + \frac{x^7}{7} - \dots$$

$$\lim_{x \rightarrow 0} \frac{\frac{x^3}{3} - \frac{x^5}{5} + \frac{x^7}{7} - \dots}{x^3} = \frac{1}{3} - \frac{x^2}{5} + \frac{x^4}{7} - \frac{x^6}{9} + \dots$$

$$\lim_{x \rightarrow 0} \frac{1}{3} - \frac{x^2}{5} + \frac{x^4}{7} - \frac{x^6}{9} + \dots = \frac{1}{3} = 0,33333...$$

№ 4: Вычислить интеграл с точностью до $\delta = 0,001$

$$\int_0^{\frac{\pi}{2}} \sqrt{x} e^{-\frac{x}{4}} dx$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$$

$$e^{-\frac{x}{4}} = 1 - \frac{x}{4 \cdot 1!} + \frac{x^2}{4^2 \cdot 2!} - \frac{x^3}{4^3 \cdot 3!} + \dots + (-1)^n \frac{x^n}{4^n \cdot n!}$$

$$\sqrt{x} e^{-\frac{x}{4}} = \sqrt{x} \left(1 - \frac{x}{4 \cdot 1!} + \frac{x^2}{4^2 \cdot 2!} - \frac{x^3}{4^3 \cdot 3!} + \dots + (-1)^n \frac{x^n}{4^n \cdot n!} \right)$$

$$\int_0^{\frac{2}{5}} \sqrt{x} e^{-\frac{x}{4}} dx = \int_0^{\frac{2}{5}} x^{\frac{1}{2}} \left(1 - \frac{x}{4 \cdot 1!} + \frac{x^2}{4^2 \cdot 2!} - \frac{x^3}{4^3 \cdot 3!} + \dots + \frac{x^n}{4^n n!} (-1)^n \right) dx = \int_0^{\frac{2}{5}} \left(x^{\frac{1}{2}} - \frac{x^{\frac{3}{2}}}{4 \cdot 1!} + \frac{x^{\frac{5}{2}}}{4^2 \cdot 2!} - \frac{x^{\frac{7}{2}}}{4^3 \cdot 3!} + \dots + (-1)^n \frac{x^{\frac{2n+1}{2}}}{4^n n!} \right) dx$$

$$= \frac{2}{3} x^{\frac{3}{2}} - \frac{2}{5} \frac{x^{\frac{5}{2}}}{4 \cdot 1!} + \frac{2}{7} \frac{x^{\frac{7}{2}}}{4^2 \cdot 2!} - \frac{2}{9} \frac{x^{\frac{9}{2}}}{4^3 \cdot 3!} + \dots + (-1)^n \frac{2}{2n+3} \frac{x^{\frac{2n+3}{2}}}{4^n n!} + \dots + C \Big|_0^{\frac{2}{5}} = \frac{2}{3} \left(\frac{2}{5} \right)^{\frac{3}{2}} - \frac{2}{5} \frac{\left(\frac{2}{5} \right)^{\frac{5}{2}}}{4 \cdot 1!} + \frac{2}{7} \frac{\left(\frac{2}{5} \right)^{\frac{7}{2}}}{4^2 \cdot 2!} - \frac{2}{9} \frac{\left(\frac{2}{5} \right)^{\frac{9}{2}}}{4^3 \cdot 3!} + \dots$$

$$\frac{2}{3} \left(\frac{2}{5} \right)^{\frac{3}{2}} = \frac{4\sqrt{10}}{75} \approx 0,168; \left| \frac{2}{5} \frac{\left(\frac{2}{5} \right)^{\frac{5}{2}}}{4 \cdot 1!} \right| = \left| -\frac{2\sqrt{10}}{625} \right| \approx 0,0101; \frac{2}{7} \frac{\left(\frac{2}{5} \right)^{\frac{7}{2}}}{4^2 \cdot 2!} = \frac{\sqrt{10}}{8750} \approx 0,000361$$

$$\int_0^{\frac{2}{5}} \sqrt{x} e^{-\frac{x}{4}} dx = \frac{4\sqrt{10}}{75} - \frac{2\sqrt{10}}{625} = 0,15853$$

№ 5

$$y' + xy = 2x^2; \quad y(0) = 0$$

$$C_0 = y = 0$$

$$y(x) = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + \dots$$

$$y'(x) = C_1 + 2C_2 x + 3C_3 x^2 + 4C_4 x^3 + 5C_5 x^4 + \dots$$

$$(C_1 + 2C_2 x + 3C_3 x^2 + 4C_4 x^3 + 5C_5 x^4 + \dots) + (C_0 x + C_1 x^2 + C_2 x^3 + C_3 x^4 + C_4 x^5 + \dots) = 2x^2$$

$$x^0: C_1 = 0$$

$$x^1: 2C_2 + C_0 = 0 \quad C_2 = 0$$

$$x^2: 3C_3 + C_1 = 2 \quad C_3 = \frac{2}{3}$$

$$x^3: 4C_4 + C_2 = 0 \quad C_4 = 0$$

$$x^4: 5C_5 + C_3 = 0 \quad C_5 = -\frac{2}{15}$$

$$x^5: 6C_6 + C_4 = 0 \quad C_6 = 0$$

$$x^6: 7C_7 + C_5 = 0 \quad C_7 = \frac{2}{105}$$

$$x^7: 8C_8 + C_6 = 0 \quad C_8 = 0$$

$$x^8: 9C_9 + C_7 = 0 \quad C_9 = -\frac{2}{945}$$

$$y = \frac{2x^2}{3} - \frac{2x^4}{15} + \frac{2x^6}{105} - \frac{2x^8}{945}$$

№ 6

$$y'' - e^y \sin y = 0; \quad y(\pi) = 1; \quad y'(\pi) = \frac{\pi}{2}; \quad y'' = e^y \sin y$$

$$y = y(\pi) + \frac{y'(\pi)}{1!} (x - \pi) + \frac{y''(\pi)}{2!} (x - \pi)^2 + \frac{y'''(\pi)}{3!} (x - \pi)^3 + \dots$$

$$1) y(\pi) = 1$$

$$2) y'(\pi) = \frac{\pi}{2}$$

$$3) y''(\pi) = e^1 \sin \frac{\pi}{2} = e$$

$$4) y'''(\pi) = e^y \sin(y) y' + e^y \cos(y) y'' = \frac{e\pi}{2}$$

$$y = 1 + \frac{\pi}{2}(x+\pi) + \frac{e}{2}(x+\pi)^2 + \frac{e\pi}{12}(x+\pi)^3$$